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University



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AMSTERDAM

Faculty of Social and Behavioral Sciences

Department of Education and Psychology

ReYoUnite: REMAIN Psychological Well-Being Survey for Refugee Youth in the Netherlands

Zainab Safar (2317435)

Internal Supervisor: Larissa Meijer

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Internship [7,5 ECTS]

Applied Cognitive Psychology

Utrecht University

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Abstract

Background:

A considerable youth number at high risk of developing mental health disorders due to the difficulties they have faced in a war zone and through a refugee journey, and in the host country, The World Health Organization (WHO) created problem management plus (pm+) as a short psychological treatment. Although PM+ does not integrate emotional processing, which is vital for those who have experienced trauma, it focuses on problem-solving, stress management, behavioural activation, and social support.

Objective:

ReYoUnite's pilot randomised controlled trial (RCT) examines the viability and level of acceptance of PM+ among youth refugees who speak Arabic, Ukrainian and Tigrinya languages and live in the Netherlands.

Methods:

Participants will be recruited by filling in a survey. Then refugee youth (N=90) between 16 and 25 years old will be into three groups randomly PM+ provided by usual care (n=30), PM+ EP provided with Emotional processing (n=30) and a control group (n=30). Non- psychology specialists and specialities (n=20) will receive PM+ training and provide it individually. The target group should report psychological distress (Kessler Psychological Distress Scale; K10 > 15) and impaired daily functioning (WHO Disability Assessment Schedule; WHODAS 2.0 > 16). The study will involve baseline assessments, as well as follow-up evaluations one week and three months after the intervention. The primary outcome of the pilot RCT is to assess the feasibility and acceptability of the adapted PM+ and PM+EP interventions. Secondary outcomes include psychological self-report distress, functional impairment, post-traumatic stress disorder (PTSD) symptom severity and diagnosis, social support, and self-identified problems.

Results and conclusion

This is the first study to assess PM+ among this age range and integrate the emotional processing module. The outcomes of this study have the potential to provide valuable insights for larger RCT and the integration of PM+ interventions for refugee youth.

Name Student: Shiyi Zhang

Thesis title: **What Mode is This? Exploring Which Elements of Visual In-vehicle HMI Designs Intuitively Indicate the Car Being in Assisted or Automated Driving Mode**

In vehicles with assisted and automated driving modes, humans and machines share the responsibilities of driving tasks and duties. However, these tasks and duties differ between modes. This means that at any moment during the journey, it should be unmistakably clear in which mode the car operates, i.e., at which level of automation. For human drivers, this is a prerequisite to be aware of who (car or human) has what responsibility. Manufacturers are increasingly developing visual Human Machine Interfaces (HMIs). This may come at the cost of consistency and compatibility between designs and could lead to mode confusion for the driver when confronted with different designs. Ideally, a certain level of standardisation is achieved amongst HMI designs to avoid mode confusion and human errors. Therefore, we 1) investigated what designs are currently being developed and 2) evaluated what elements affect the driver's intuitive decision on mode status.

To obtain an overview of the existing designs, desk research was performed, and car manufacturers were asked to share their HMI designs for assisted and automated driving modes. Commonalities and differences between designs were identified. Next, in an experimental study, different visual designs were tested to assess whether they are perceived as assisted or automated driving modes. First, to assess intuitive associations, participants were asked to indicate the automation level corresponding to the HMI shown as quickly as possible. For this, reaction time and response accuracy were measured. Through subsequent guided interviews, insight was obtained into which elements aided decision-making.

This research provides the state of the art regarding current HMIs for assisted and automated driving and identifies elements that affect drivers' intuitive decisions on mode status. Based on these findings, solutions are proposed to tackle issues in HMI design related to mode confusion and to provide input for Euro NCAP protocols.

Name Company and University: RDW & Utrecht University

External supervisor: Ilse Harms

UU Supervisor: Stella Donker

Name Student: Pedro José Trueba

Internship or Thesis: Combined

Title project: Comparing Self-Reports and Hidden Markov Modelling to Understand Strategies in Economic Decision-Making.

Abstract:

When individuals respond to a task, there are multiple strategies they can use, but it's not possible to access their experiences directly. Therefore, different methods have been used to discover these strategies. Self-reports and Hidden Markov Modelling are two examples of such methods. However, self-reports have been criticized because people might not be able to accurately report what they did, while Hidden Markov Modelling assumes that different behaviors correspond to different strategies, which may not always be true. To understand different strategies in economic decision-making, this project aims to compare these two methods and determine their convergence or divergence. A set of design requirements for an experimental setup is necessary to answer these questions.

Name company or University: Utrecht University

UU Supervisor: Leendert van Maanen

Name Student: Sophia Wantenaar

Internship or Thesis: Thesis

Title project: Push me to the (V)Edge: using Emergency Reserves for Goal Persistence after Subgoal Failure.

The process of goal pursuit is often subject to various external, interrupting factors causing people to disengage from their goals after subgoal failure. This study looks into the possibility of extending existing research to a dietary context, particularly, consuming higher shares of vegetarian meals, as this benefits health and the environment. In addition, this research looks into whether the mechanism behind emergency reserves can be explained similarly to that of flexible goal settings. Therefore, feelings of accomplishment are tested as a mediator. For this five-week research participants filled out Google Spreadsheets with a dietary diary as well as weekly questionnaires regarding feelings of accomplishment. No significant results were found regarding the hypotheses. However, there was a significant effect of feelings of accomplishment on goal persistence. One interpretation is that goal framing using emergency reserves cannot be explained similarly to flexible goal setting.

Name company or University: Utrecht University

UU Supervisor: Robert Weijers

Name Student: Surya Tjio

Internship or Thesis: Thesis

Title project: "Fact or Fad: A self-paced reading study on the effect of Bionic Reading on reading speed and comprehension in Dutch"

Research shows that increased reading speeds are unlikely to result in equal or increased comprehension of a text. There is a trade-off between these factors. Bionic Reading is a new method that aims to increase reading speed and enable a deeper understanding of a text through its unique typeface design. Using two Self-Paced reading tasks, this thesis compared Bionic Reading with a standard typeface to look at its effect on reading speed and comprehension. Task 1 focused on reading speed and global-level comprehension by presenting paragraphs on screen. Task 2 focused on sentence-level semantic processing by presenting sentences word by word. No significant differences were found between Bionic Reading and the standard typeface. However, limitations in this study make it difficult to draw definite conclusions.

Name company or University: Utrecht University

(If applicable) External supervisor: N/A

UU Supervisor: Suzanne Hut

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11-04-2023

Name Student: Julia Schout

Internship or Thesis: Internship

Name company or University: KRNWTR+

(If applicable) External supervisor: Danielle van Wessem

UU Supervisor: Surya Gayet

Title: Sipping Sustainability: A cognitive psychological approach to developing marketing materials
for KRNWTR+

This internship report reports on the challenge of effectively communicating sustainability in businesses, with a specific focus on the case of KRNWTR+. As a social enterprise that produces sustainable water dispensers to reduce the waste of disposable plastic water bottles, KRNWTR+ is an illustrative example of a business focusing on sustainability. The report highlights the risks associated with information overload in sustainability communication, including green fatigue and cynicism. I, as an Applied Cognitive Psychology, have aimed to develop marketing materials that target individual consumers by understanding their needs, behavior, and decision-making processes. These materials are designed with low cognitive load, aesthetic appeal, easy manufacturability, and a transparent mission in mind. The conclusion emphasizes the importance of genuine concern for sustainability, the avoidance of information overload, and meaningful audience engagement. The report acknowledges limitations and suggests the need for further research while remaining optimistic about the potential effectiveness of the developed materials in promoting a sustainable lifestyle beyond just environmental sustainability.

Name Student: Arslaan Amjad

Internship or Thesis: Internship

Title project: Detecting visual deficits.

Detecting visual field deficits at an early age is imperative as it can prevent any visual deterioration. However, assessing the visual field of children remains a challenging procedure as these require a certain degree of cognitive capacity and (non)verbal skills, which young children might not possess yet. This may pose a challenge when working with young children when trying to diagnose them with a visual deficit. It is therefore important to develop an objective tool that gives reliable results in the face of these challenges. Measuring the pupillary response may provide a solution for these challenges. The aim of this research will be to, first, explore whether pupil responses in the face of changes in color in the environment are indicative of color blindness, and second, whether the severity of color blindness can be determined when comparing these pupil size oscillations in a standardized environment. It is expected that people with color blindness will have weaker response to the changing colors than people who are not colorblind.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Marnix Naber

Name Student: Emmie van Zijtveld

Internship or Thesis: Internship

Title project: Improving language proficiency: The effect of practicing in Hogeschooltaal on the passing rate of Dutch college students on a 3F technical writing skills test.

This research investigates the effect of practicing Hogeschooltaal, an online learning platform for college students for the Dutch and English writing skills, on the technical writing skills of Dutch college students. Data on test scores and practicing behavior were gathered from Hogeschooltaal's database. The focus of this research is on Dutch writing skills. Descriptive statistics, chi-squared tests and visualizations of the data were used for analysis. First of all only 25% of all students passed the intake test on 3F level, showing a lack in current writing skills. In addition to this, this research shows a significant relationship between exercises done in Hogeschooltaal and the chances of passing a technical writing multiple choice test. This suggests that practicing in Hogeschooltaal is effective for Dutch college students to gain better technical writing skills. This is promising for Hogeschooltaal, however more research is needed into other factors that influence learning behavior.

Name company or University: Noordhoff Uitgevers

(If applicable) External supervisor: Eva Landman

UU Supervisor: Suzanne Hut

Name Student: Nikita V. Kondratyev

Internship or Thesis: Thesis

Title project: Psychophysiological responses to threats of differing probabilities

For my thesis, I am working alongside another master's student in the neuroscience program, and together we are designing an experiment that aims to improve on past studies that have tested the threat probability/NPU paradigm. The main study we are improving on is Bradford et al's 2014 study which has some potentially confounding variables, and only measures startle response. So our study will adjust their design to remove those potential confounds and expand on the psychophysiological measures taken from just startle, to startle, postural sway/freezing, EEG measures, heart rate, and skin conductance response. Our study has not begun data collection yet, thus we can not make any conclusions yet, but I will be happy to introduce our experimental design and discuss the literature that has informed our design

Name company or University: Utrecht University

(If applicable) External supervisor: N/A

UU Supervisor: Joke Baas

Name Student: Po-Hsuan, Huang

Internship or Thesis: Thesis(27.5)

Title project:

The real the better? Comparing people's differences in reacting to photos and images from high to low detail.

Abstract

The idea of this research project is inspired by a Dutch company transferring traditional word-based warning signs in the working areas into photorealistic images as a visual language.

In this experiment, we mainly focus on participants' picture recognition performance of images with different levels of detail. We create the stimulus by manipulating the photos' posterization level, which is one of the key differences between photos and illustrations.

The experiment consists of two main phases: Study and Test. First, we will randomly present images (photos or photorealistic images) to the participants. Then, in the test phase, we will display images to the participants, and they will have to answer whether the images were displayed in the previous phase.

The experiment is based on Pychopy and is currently under construction. However, we expect to start it shortly.

Name company or University: Utrecht University

UU Supervisor: Dr. Suzanne Hut

Kallia Karioti (4431391)

Thesis project

Mixed Reality in Cognitive Rehabilitation

(Abstract, 100-150 words)

Utrecht University

UU Supervisor: Charlotte van de Wouw

The title of this thesis project is “Mixed Reality in cognitive rehabilitation”. Its main goal is to find out the importance of MR in cognitive rehabilitation, specifically cognitive assessment, and its potential advantages and disadvantages. Moreover, in order to find out what exactly MR provides to therapy, is necessary to gain insight into the potential different perceptions of healthcare professionals (e.g., clinicians and researchers) towards MR in cognitive rehabilitation and whether they have an experience using MR during their career. Taking all the above into consideration, through this research many different definitions and perceptions will be analyzed. The main research questions are the following: “Which are the potential differences between disciplines working in the field of cognitive rehabilitation?” and “What are the Clinician’s and researcher’s perspectives regarding potential strengths and weaknesses of MR in cognitive assessment?”. An online survey is the research tool which is used for this project. This survey consists of 20 questions, about the potential experience that the healthcare professionals have with Mixed Reality, and how useful or not they think Mixed Reality is in cognitive rehabilitation. There are almost 50 respondents, and their answers were recorded through Qualtrics. Lastly, the collected data will be analyzed through SPSS.

Name Student: Mark van Ditmarsch

Internship or Thesis: Thesis

Title project: The Effect of Bionic Reading on Semantic Processing during Speed-Reading.

Previous research has proven a trade-off where an increased reading speed decreases reading comprehension. This study aimed to test whether the Bionic Reading (BR) font could help people retain reading comprehension, in the form of semantic processing, while speed-reading at a rate that is likely to cause a decrease in comprehension in a standard font. A RSVP-task setup was used to record EEG while participants were shown congruent and incongruent sentences at a rate of 500 words per minute. The findings indicated a significant N400 effect between the congruent and incongruent sentences, implying that semantic processing was not impaired by the used word rate. This effect could be attributed by the selected interstimulus interval (ISI). Due to limitations with the ISI, no conclusion could be drawn regarding the effect of BR during speed-reading.

Name company or University: Utrecht University

UU Supervisor: S.C.A. Hut

Name Student: Loes Brust

Internship or Thesis: marketing intern at REVIT! Sport

Title project: The sustainable drivers of motorcyclists and their motivations for riding.

REV'IT! is a motorcycle clothing company that is interested in communicating their sustainable development and looking to inspire others to join. Sustainable development captures the need to preserve natural resources, biodiversity, human welfare, and equity. To find the most suited communication strategy, literature research is done to get a better understanding of current communication trends and their effectiveness on personality types. Besides, a questionnaire is performed to garner the specifications and personality types of the consumers and their interest in sustainable development. Conform hypotheses, personality types were correlated to the rider motivations, as well as partly to sustainable attitudes. With this knowledge, riders with different rider motivations and personalities can be approached in most suited manner regarding sustainable communication. These findings give more insight into the influence of personality on sustainable attitude, as well as offers proposed communication strategies based on these differences.

Name company or University: REV'IT! Sport International

(If applicable) External supervisor: Iain Howe, Global Marketing Director

UU Supervisor: Stella Donker

Updatedag 2 Samuel van der Loo

Name Student: Samuel van der Loo

Internship: UX-design

Titel project: Banking for better customer service, for generations to come.

Abstract

Currently, I am doing an internship at ABN AMRO. The position is UX design in the Customer Interaction department. For my research, I take a deeper look into the interpretation of the waiting time of customer service on the phone. Complaints have been received by customers about this interpretation. Currently, there is music playing while waiting. The customers found this too unpleasant. Can this be improved? The research question is: *which sound interpretation during the waiting time on the phone of a customer service of a bank increases the patience of the customer?* Here I try to find out, through a survey and prototype testing, which interpretation promotes the customer's patience. The aim is to promote the patience of the clients and thus improve the experience with ABN AMRO.

Name company: ABN AMRO Group N.V.

External supervisor: Femke van Drooge

UU supervisor: Sam van Bijnen

Sofia Balali

Internship

Title project: Why do Conspiracy Theories exist? Evolution, Cognitive Mechanisms, and Motives: A Literature Review

TNO, Soesterberg

External Supervisor : Hans Korteling (hans.korteling@tno.nl)

UU Supervisor : Suzanne Hut (s.c.a.hut@uu.nl)

Belief in conspiracy theories(CTs) is a widespread phenomenon that is time and context-independent that impacts negatively both individuals and societies. This paper investigates the tendencies in humans, such as evolutionary, neural, and cognitive, that render the individual prone to conspiratorial thinking and the motives behind the endorsement and dissemination of CTs on individual and group level. Prior research has investigated CTs mostly from the perspective of personality traits, epistemic deficiencies, and political implications. However, an inclusive framework addressing other variables is lacking. Results have shown that during our evolution, conspiratorial thinking tendencies developed to maximize survival and fitness and were possibly strengthened by the underlying neural networks(association, compatibility etc.) and the resulting cognitive mechanisms, such as biases. In addition, endorsement and dissemination of CTs can be influenced greatly by external factors such as motivation, political, and socioeconomic situation. This review will propose a more inclusive framework that can be used to inform future interventions to mitigate the effects of conspiracy theories.

Keywords: conspiracy theories, cognitive biases, heuristics, evolution, motives, neural network

Name Student: Shawn Veldhuis

Internship or Thesis: Thesis

Title project: How Trust Affects Pedestrian Behavior around Autonomous Vehicles presenting Incongruent HMI: A Comparative Study of Left- and Right-Traffic Countries.
(Abstract, 100-150 words)

Name company or University: German Aerospace Center (DLR)

(If applicable) External supervisor:

UU Supervisor: Chris Janssen

Establishing a set of rules and guidelines for the use of explicit communication by autonomous vehicles (AVs) through external human machine interfaces (eHMI) could make AVs more reliable and feasible for society. To achieve this, it is essential to investigate whether the proposed 360° light-band on different vehicle sizes is universally understood by pedestrians from different countries. Additionally, there needs to be a level of trust between pedestrians and AVs eHMI for it to fulfill its purpose. Overtrust in the eHMI could lead to safety issues and decrease trust in AVs. This study examines the impact of pedestrians' initial trust on the willingness to cross when the AV shows conflicting behavior between its explicit and implicit communication. The study compares left-way traffic and right-way traffic countries, including the UK and The Netherlands, by analyzing data from an online questionnaire. The results show that the proposed eHMI is universally understood, but it leads to overtrust in participants with high trust in AVs. This highlights the need to find ways to increase safety and reliance in AVs without fostering overtrust in participants.

Name Student: Carmen van Klaren

Internship or Thesis: Thesis

Title project: The validity of cEEGrids (ear-EEG) for fatigue detection in aviation

Fatigue is a major concern in safety-critical industries like aviation as it can result in errors and accidents. To mitigate this risk, non-invasive methods to monitor fatigue are needed. A new approach involves using electroencephalography (EEG) integrated into headphones, which can measure EEG oscillations associated with fatigue. This study aimed to validate the use of cEEGrids, a type of ear-EEG, for measuring fatigue. The results showed that cEEGrids were a valid measure of fatigue, with participants who experienced sleep restriction exhibiting a significant increase in relative alpha band power in some channels. This increase correlated with both subjective KSS scores and response speed in the PVT task. However, the effects were limited to certain channels and participants experienced only moderate sleep restriction. Further research should optimize EEG signals and channel locations, as well as examine the effects of extreme sleep deprivation. Nonetheless, the study highlights the potential of ear-EEG for live monitoring and comprehensive research on fatigue in aviation.

Name company or University: Royal Netherlands Aerospace Centre (NLR)

External supervisor: Alwin van Drongelen

UU Supervisor: Robert Weijers

Name student: Gu Zirui

Internship or thesis: internship

Title project: A Pilot Study of the Development of a Test to Differentiate Subtypes of Hemispatial Neglect: Egocentric vs Allocentric Neglect

Egocentric neglect (view-based bias) and allocentric neglect (object-based bias) are two subtypes of hemispatial neglect and they are mostly differentiated by paper and pencil tasks. However, these tests require overt responses, whereby a response may affect measurements. Research by Strauch et al. (2022) indicated that pupil light responses reveal pseudoneglect and offer the possibility for the development of new and objective diagnostic tests for hemispatial neglect. The aim of this study is the development of a pupillometric test battery for differentiating egocentric and allocentric neglect. A pilot study was conducted with 33 healthy participants, all of whom were instructed to complete a pupillometry test where six stimuli were randomly presented. The results demonstrate that in the comparison of Black/White_Full vs White_Full and Black/White_One vs White/Black_One, pupil light responses do not indicate left bias in healthy control, contrary to our hypothesis. Instead, pupil constriction is found to be significantly larger when bars were not connected, which shown to be two bars, thereby indicating leftward attention bias. As leftward attentional bias was only found in one of the three stimuli layouts, it remains to be tested whether this method can indeed differentiate egocentric and allocentric neglect. Further research is required, and these methods must be retested in both healthy control and egocentric/allocentric neglect patients.

Name company or university: Utrecht University

UU supervisor: Christoph Strauch

Name **Student:** Fernando López Rodríguez-Carretero
Internship **or** **Thesis:** Thesis

Title project: Using nudges to detect gender bias in job advertisements: an experimental study to analyse job recruiters' perception of hiring in diversity (draft title).

** Given that the study is still in the data collection phase and, therefore, has not yet been completed, the text presented here does not follow the abstract structure. However, it gives a clear idea of what the study consists of. This text has been included at the end of the participants' survey to inform them about the study.*

The study behind this survey is focused on identifying whether gendered wording in job advertisements influences your perception as if you were job recruiters, of deciding about hiring a man or a woman for the specific job you read. It is hypothesised that gendered wording interacts with the job recruiters' perception of job advertisements. This could maintain a direct effect on the diversity within the recruitment process.

Each advertisement used specific female or male wording within traditionally female-dominated or male-dominated occupations, thereby getting two different configurations of the same job advertisement. This is known as gender reframing. The latter aimed to influence your perception of the job and whether it should be more convenient to be performed by a man or a woman. By answering the questions you encountered, you were deciding about hiring potentially a woman or a man for the specific job based on the impression after reading the job advertisement.

In general terms, this study could have important implications for intervening in the existing situation of gender inequality in recruitment processes along with the evident unbalanced access to job opportunities within the labour market, serving as a preliminary diagnostic index for the study of the current state of the situation, as well as to set the stage for the design of protocol measures to guide recruitment processes systemically in all workplaces.

Name **company** **or** **University:** Universiteit Utrecht
(If **applicable)** **External** **supervisor:**
UU Supervisor: Robert Weijers

Name: Michail Olympios

Internship

Title: Measuring heart rate using a webcam: Employing remote photoplethysmography to map vascular activity of face skin tissues during demanding tasks

Objective: Experimentally test and measure the effect of remote photoplethysmography on facial skin pulsation and heart rate during mentally and physically demanding tasks

Background: Remote photoplethysmography (rPPG) allows contact-free tracking of an individual's heart rate, blood pressure and cardiovascular functionality via a digital camera. Previous work focused on the construction of supervised and unsupervised algorithms and their reliability as well as testing them on different situations such as skin tone, physical exercise, and sleep, mainly with the metrics of blood volume pressure, heart rate, and heart rate variability. In this study, rPPG is utilized in a combination of cognitive, physical stressors, and rest estimating how pulsatile is the face, the variations of heart rate, and signal-to-noise ratio values during these conditions. To our knowledge, this is the first investigation evaluating the pulsatility of the face during a variety of demanding tasks. Two state-of-the-art algorithms, plane orthogonal-to-skin (POS) and independent component analysis – Naber (ICA) algorithms were performed while a commercial oximeter acted as a reference to the measurements.

Methods: Seven healthy participants were recruited, and face recordings were made accordingly. Participants had to perform three mentally demanding tasks, the Stroop task, Arithmetic task and Pitch task as well as two physically demanding tasks, a light and medium-level burpee exercise. Every participant had to stand in front of a black wall looking at the camera while performing each task independently.

Results: We report a moderate accuracy of the two algorithms' aimed at the facial skin tissues in response to ground truth, the oximeter. There is a direction to higher face skin pulsation, blood flow during cognitive stressors than physical exercise and vice versa. Heart rate is increased during physical exercise than cognitive stressors and baseline measurements. The two latter, baseline and cognitive stressors heart rate remained relevantly the same. Baseline SNR are significantly higher than the two types of stressors, pointing out the involvement of activity. However, low SNR values were extracted deteriorating the quality of signal.

Conclusion: rPPG shows potential indications to capture the pulsatility of the face and blood flow during various demanding tasks as well as the heart rate.

UU supervisor: Marnix Naber

Psychophysiology of Perception Laboratory

Experimental Psychology Department

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